

Questions

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1. Given that

$$y = \arctan\left(\frac{x-2}{x+2}\right)$$

where $x \neq -2$

(a) find $\frac{dy}{dx}$, giving your answer in its simplest form. [2]

(b) Use integration by parts to find

$$\int \arctan\left(\frac{x-2}{x+2}\right) dx \quad [4]$$

Solution

(a) Let

$$u = \frac{x-2}{x+2}$$

so that

$$y = \arctan u$$

Using the chain rule,

$$\frac{dy}{dx} = \frac{1}{1+u^2} \cdot \frac{du}{dx}$$

Now differentiate u :

$$\begin{aligned} \frac{du}{dx} &= \frac{(x+2)(1) - (x-2)(1)}{(x+2)^2} \\ &= \frac{x+2-x+2}{(x+2)^2} \\ &= \frac{4}{(x+2)^2} \end{aligned}$$

Hence

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{1 + \left(\frac{x-2}{x+2}\right)^2} \cdot \frac{4}{(x+2)^2} \\ &= \frac{4}{(x+2)^2 + (x-2)^2} \\ &= \frac{4}{(x^2 + 4x + 4) + (x^2 - 4x + 4)} \\ &= \frac{4}{2x^2 + 8} \\ &= \frac{2}{x^2 + 4} \end{aligned}$$

Therefore,

$$\frac{dy}{dx} = \frac{2}{x^2 + 4}$$

(b) Let

$$I = \int \arctan\left(\frac{x-2}{x+2}\right) dx$$

Use integration by parts with

$$u = \arctan\left(\frac{x-2}{x+2}\right), \quad \frac{dv}{dx} = 1$$

Then

$$v = x \quad \text{and} \quad \frac{du}{dx} = \frac{2}{x^2 + 4}$$

from part (a).

Using

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

gives

$$\begin{aligned} I &= x \arctan\left(\frac{x-2}{x+2}\right) - \int x \cdot \frac{2}{x^2+4} dx \\ &= x \arctan\left(\frac{x-2}{x+2}\right) - \int \frac{2x}{x^2+4} dx \end{aligned}$$

To evaluate the remaining integral, let

$$t = x^2 + 4$$

so that

$$\frac{dt}{dx} = 2x \quad \Rightarrow \quad dt = 2x dx$$

Therefore

$$\begin{aligned} \int \frac{2x}{x^2+4} dx &= \int \frac{1}{t} dt \\ &= \ln t + C \\ &= \ln(x^2+4) + C \end{aligned}$$

So

$$I = x \arctan\left(\frac{x-2}{x+2}\right) - \ln(x^2+4) + C$$

Hence,

$$\int \arctan\left(\frac{x-2}{x+2}\right) dx = x \arctan\left(\frac{x-2}{x+2}\right) - \ln(x^2+4) + C$$

2. Use integration by parts to find

$$\int_0^2 x^2 \ln(1+x) \, dx$$

giving your answer in the form $a + b \ln 3$, where a and b are constants to be found.

[5]

Solution

Let

$$I = \int_0^2 x^2 \ln(1+x) \, dx$$

Use integration by parts with

$$u = \ln(1+x), \quad \frac{dv}{dx} = x^2$$

so that

$$\frac{du}{dx} = \frac{1}{1+x}, \quad v = \frac{x^3}{3}$$

Then

$$\begin{aligned} I &= \left[\frac{x^3}{3} \ln(1+x) \right]_0^2 - \int_0^2 \frac{x^3}{3(1+x)} \, dx \\ &= \left[\frac{x^3}{3} \ln(1+x) \right]_0^2 - \frac{1}{3} \int_0^2 \frac{x^3}{1+x} \, dx \end{aligned}$$

Now simplify the fraction:

$$\frac{x^3}{1+x} = x^2 - x + 1 - \frac{1}{1+x}$$

since

$$(1+x)(x^2 - x + 1) = x^3 + 1$$

So

$$\begin{aligned} \int_0^2 \frac{x^3}{1+x} \, dx &= \int_0^2 \left(x^2 - x + 1 - \frac{1}{1+x} \right) \, dx \\ &= \left[\frac{x^3}{3} - \frac{x^2}{2} + x - \ln(1+x) \right]_0^2 \\ &= \left(\frac{8}{3} - 2 + 2 - \ln 3 \right) - (0 - 0 + 0 - \ln 1) \\ &= \frac{8}{3} - \ln 3 \end{aligned}$$

Also,

$$\left[\frac{x^3}{3} \ln(1+x) \right]_0^2 = \frac{8}{3} \ln 3$$

Hence

$$\begin{aligned} I &= \frac{8}{3} \ln 3 - \frac{1}{3} \left(\frac{8}{3} - \ln 3 \right) \\ &= \frac{8}{3} \ln 3 - \frac{8}{9} + \frac{1}{3} \ln 3 \\ &= 3 \ln 3 - \frac{8}{9} \end{aligned}$$

Therefore

$$\int_0^2 x^2 \ln(1+x) \, dx = -\frac{8}{9} + 3 \ln 3$$

So

$$a = -\frac{8}{9}, \quad b = 3$$

3. Use integration by parts to find

$$\int \operatorname{arsinh}(2x + 1) \, dx$$

giving your answer in terms of logarithms.

[5]

Solution

Let

$$I = \int \operatorname{arsinh}(2x + 1) \, dx$$

Use the substitution

$$t = 2x + 1 \quad \Rightarrow \quad dx = \frac{1}{2} dt$$

Then

$$I = \frac{1}{2} \int \operatorname{arsinh}(t) \, dt$$

Now integrate by parts. Take

$$U = \operatorname{arsinh}(t), \quad \frac{dV}{dt} = 1$$

so that

$$\frac{dU}{dt} = \frac{1}{\sqrt{t^2 + 1}}, \quad V = t$$

Using

$$\int U \frac{dV}{dt} dt = UV - \int V \frac{dU}{dt} dt$$

gives

$$\int \operatorname{arsinh}(t) \, dt = t \operatorname{arsinh}(t) - \int \frac{t}{\sqrt{t^2 + 1}} \, dt$$

Now

$$\frac{d}{dt}(\sqrt{t^2 + 1}) = \frac{t}{\sqrt{t^2 + 1}}$$

so

$$\int \frac{t}{\sqrt{t^2 + 1}} \, dt = \sqrt{t^2 + 1}$$

Hence

$$\int \operatorname{arsinh}(t) \, dt = t \operatorname{arsinh}(t) - \sqrt{t^2 + 1} + C$$

Therefore

$$I = \frac{1}{2} \left(t \operatorname{arsinh}(t) - \sqrt{t^2 + 1} \right) + C$$

Substituting back $t = 2x + 1$,

$$I = \frac{1}{2} \left((2x + 1) \operatorname{arsinh}(2x + 1) - \sqrt{(2x + 1)^2 + 1} \right) + C$$

Finally, using

$$\operatorname{arsinh}(y) = \ln(y + \sqrt{y^2 + 1})$$

we get

$$\int \operatorname{arsinh}(2x + 1) \, dx = \frac{1}{2} \left((2x + 1) \ln(2x + 1 + \sqrt{(2x + 1)^2 + 1}) - \sqrt{(2x + 1)^2 + 1} \right) + C$$

So the answer in logarithmic form is

$$\frac{1}{2} \left((2x + 1) \ln(2x + 1 + \sqrt{(2x + 1)^2 + 1}) - \sqrt{(2x + 1)^2 + 1} \right) + C$$

4.

$$h(\theta) = (3 \sin \theta - \cos \theta)(\sin \theta + 2 \cos \theta)$$

(a) Show that $h(\theta) = \frac{1}{2} - \frac{5}{2} \cos 2\theta + \frac{5}{2} \sin 2\theta$. [2]

(b) Hence, using calculus, find the exact value of

$$\int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \theta \right) h(\theta) \, d\theta$$
 [5]

Solution

(a) Expanding the brackets,

$$\begin{aligned} h(\theta) &= (3 \sin \theta - \cos \theta)(\sin \theta + 2 \cos \theta) \\ &= 3 \sin^2 \theta + 6 \sin \theta \cos \theta - \sin \theta \cos \theta - 2 \cos^2 \theta \\ &= 3 \sin^2 \theta + 5 \sin \theta \cos \theta - 2 \cos^2 \theta \end{aligned}$$

Now use the double-angle identities

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}, \quad \cos^2 \theta = \frac{1 + \cos 2\theta}{2}, \quad \sin \theta \cos \theta = \frac{1}{2} \sin 2\theta$$

Substituting,

$$\begin{aligned} h(\theta) &= 3 \left(\frac{1 - \cos 2\theta}{2} \right) + 5 \left(\frac{1}{2} \sin 2\theta \right) - 2 \left(\frac{1 + \cos 2\theta}{2} \right) \\ &= \frac{3}{2} - \frac{3}{2} \cos 2\theta + \frac{5}{2} \sin 2\theta - 1 - \cos 2\theta \\ &= \frac{1}{2} - \frac{5}{2} \cos 2\theta + \frac{5}{2} \sin 2\theta \end{aligned}$$

Hence,

$$h(\theta) = \frac{1}{2} - \frac{5}{2} \cos 2\theta + \frac{5}{2} \sin 2\theta$$

(b) Let

$$I = \int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \theta \right) h(\theta) \, d\theta$$

Using part (a),

$$\begin{aligned} I &= \int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \theta \right) \left(\frac{1}{2} - \frac{5}{2} \cos 2\theta + \frac{5}{2} \sin 2\theta \right) \, d\theta \\ &= \frac{1}{2} \int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \theta \right) \, d\theta - \frac{5}{2} \int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \theta \right) \cos 2\theta \, d\theta + \frac{5}{2} \int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \theta \right) \sin 2\theta \, d\theta \end{aligned}$$

We evaluate the three integrals separately.

First,

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \theta \right) \, d\theta &= \left[\frac{\pi\theta}{2} - \frac{\theta^2}{2} \right]_0^{\frac{\pi}{2}} \\ &= \frac{\pi^2}{4} - \frac{\pi^2}{8} \\ &= \frac{\pi^2}{8} \end{aligned}$$

So the first term contributes

$$\frac{1}{2} \cdot \frac{\pi^2}{8} = \frac{\pi^2}{16}$$

Now let

$$B = \int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \theta \right) \cos 2\theta \, d\theta$$

Integrating by parts with

$$u = \frac{\pi}{2} - \theta, \quad \frac{dv}{d\theta} = \cos 2\theta$$

gives

$$\frac{du}{d\theta} = -1, \quad v = \frac{1}{2} \sin 2\theta$$

Hence,

$$\begin{aligned} B &= \left(\frac{\pi}{2} - \theta \right) \left(\frac{1}{2} \sin 2\theta \right) \Big|_0^{\frac{\pi}{2}} + \frac{1}{2} \int_0^{\frac{\pi}{2}} \sin 2\theta \, d\theta \\ &= 0 + \frac{1}{2} \left[-\frac{1}{2} \cos 2\theta \right]_0^{\frac{\pi}{2}} \\ &= \frac{1}{2} \left(-\frac{1}{2} \cos \pi + \frac{1}{2} \cos 0 \right) \\ &= \frac{1}{2} \left(\frac{1}{2} + \frac{1}{2} \right) \\ &= \frac{1}{2} \end{aligned}$$

Next let

$$C = \int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \theta \right) \sin 2\theta \, d\theta$$

Again integrating by parts with

$$u = \frac{\pi}{2} - \theta, \quad \frac{dv}{d\theta} = \sin 2\theta$$

gives

$$\frac{du}{d\theta} = -1, \quad v = -\frac{1}{2} \cos 2\theta$$

So,

$$\begin{aligned} C &= - \left(\frac{\pi}{2} - \theta \right) \left(\frac{1}{2} \cos 2\theta \right) \Big|_0^{\frac{\pi}{2}} - \frac{1}{2} \int_0^{\frac{\pi}{2}} \cos 2\theta \, d\theta \\ &= - \left[\left(\frac{\pi}{2} - \theta \right) \frac{1}{2} \cos 2\theta \right]_0^{\frac{\pi}{2}} - \frac{1}{2} \left[\frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{2}} \\ &= \frac{\pi}{4} - 0 \\ &= \frac{\pi}{4} \end{aligned}$$

Substituting these into the expression for I ,

$$\begin{aligned} I &= \frac{\pi^2}{16} - \frac{5}{2} \left(\frac{1}{2} \right) + \frac{5}{2} \left(\frac{\pi}{4} \right) \\ &= \frac{\pi^2}{16} - \frac{5}{4} + \frac{5\pi}{8} \\ &= \frac{\pi^2 + 10\pi - 20}{16} \end{aligned}$$

Therefore, the exact value is

$$\int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \theta \right) h(\theta) \, d\theta = \frac{\pi^2 + 10\pi - 20}{16}$$

This is approximately 1.33.

5. (a) Find

$$\int x^2 \arctan x \, dx \quad [5]$$

(b) Hence find the exact value of

$$\int_0^1 x^2 \arctan x \, dx \quad [2]$$

Solution

(a) Let

$$I = \int x^2 \arctan x \, dx$$

Use integration by parts with

$$u = \arctan x, \quad \frac{dv}{dx} = x^2$$

so that

$$\frac{du}{dx} = \frac{1}{1+x^2}, \quad v = \frac{x^3}{3}$$

Then

$$\begin{aligned} I &= uv - \int v \frac{du}{dx} \, dx \\ &= \frac{x^3}{3} \arctan x - \int \frac{x^3}{3(1+x^2)} \, dx \\ &= \frac{x^3}{3} \arctan x - \frac{1}{3} \int \frac{x^3}{1+x^2} \, dx \end{aligned}$$

Now simplify the fraction:

$$\frac{x^3}{1+x^2} = x - \frac{x}{1+x^2}$$

since

$$x - \frac{x}{1+x^2} = \frac{x(1+x^2) - x}{1+x^2} = \frac{x^3}{1+x^2}$$

So

$$\begin{aligned} I &= \frac{x^3}{3} \arctan x - \frac{1}{3} \int \left(x - \frac{x}{1+x^2} \right) \, dx \\ &= \frac{x^3}{3} \arctan x - \frac{1}{3} \left(\int x \, dx - \int \frac{x}{1+x^2} \, dx \right) \end{aligned}$$

Now

$$\int x \, dx = \frac{x^2}{2}$$

and, using the substitution $1+x^2$,

$$\int \frac{x}{1+x^2} \, dx = \frac{1}{2} \ln(1+x^2)$$

Hence

$$\begin{aligned} I &= \frac{x^3}{3} \arctan x - \frac{1}{3} \left(\frac{x^2}{2} - \frac{1}{2} \ln(1+x^2) \right) + C \\ &= \frac{x^3}{3} \arctan x - \frac{x^2}{6} + \frac{1}{6} \ln(1+x^2) + C \end{aligned}$$

Therefore,

$$\int x^2 \arctan x \, dx = \frac{x^3}{3} \arctan x - \frac{x^2}{6} + \frac{1}{6} \ln(1+x^2) + C$$

(b) Using the result from part (a),

$$\int_0^1 x^2 \arctan x \, dx = \left[\frac{x^3}{3} \arctan x - \frac{x^2}{6} + \frac{1}{6} \ln(1+x^2) \right]_0^1$$

Substituting the limits:

$$\begin{aligned} \int_0^1 x^2 \arctan x \, dx &= \left(\frac{1}{3} \arctan 1 - \frac{1}{6} + \frac{1}{6} \ln 2 \right) - \left(0 - 0 + \frac{1}{6} \ln 1 \right) \\ &= \frac{1}{3} \cdot \frac{\pi}{4} - \frac{1}{6} + \frac{1}{6} \ln 2 \end{aligned}$$

Since $\ln 1 = 0$,

$$\int_0^1 x^2 \arctan x \, dx = \frac{\pi}{12} - \frac{1}{6} + \frac{\ln 2}{6}$$

So the exact value is

$$\frac{\pi + 2 \ln 2 - 2}{12}$$

6. (i) Sketch the graph of $y = \arctan x$ for $x \in \mathbb{R}$, indicating any horizontal asymptotes. [1]

(ii) Prove that

$$\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2} \quad [4]$$

(iii) Using integration by parts and suitable substitutions, show that

$$\int_0^1 x(\arctan x)^2 dx = \frac{\pi^2}{16} - \frac{\pi}{4} + \frac{1}{2} \ln 2 \quad [6]$$

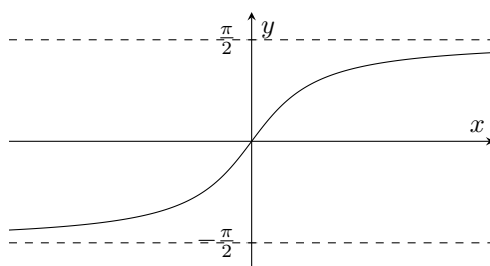
Solution

(i) The graph of $y = \arctan x$ is an increasing odd curve passing through $(0, 0)$.

As $x \rightarrow \infty$, $y \rightarrow \frac{\pi}{2}$, and as $x \rightarrow -\infty$, $y \rightarrow -\frac{\pi}{2}$.

So the horizontal asymptotes are

$$y = \frac{\pi}{2} \quad \text{and} \quad y = -\frac{\pi}{2}$$



(ii) Let

$$y = \arctan x$$

Then

$$x = \tan y$$

Differentiate implicitly with respect to x :

$$1 = \sec^2 y \frac{dy}{dx}$$

So

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{\sec^2 y} \\ &= \frac{1}{1 + \tan^2 y} \\ &= \frac{1}{1 + x^2} \end{aligned}$$

since $\tan y = x$.

Therefore

$$\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$$

(iii) Let

$$I = \int_0^1 x(\arctan x)^2 dx$$

Use integration by parts with

$$u = (\arctan x)^2, \quad \frac{dv}{dx} = x$$

Then

$$\frac{du}{dx} = 2(\arctan x) \cdot \frac{1}{1+x^2}, \quad v = \frac{x^2}{2}$$

Hence

$$\begin{aligned} I &= \left[\frac{x^2}{2} (\arctan x)^2 \right]_0^1 - \int_0^1 \frac{x^2}{2} \cdot 2 \frac{\arctan x}{1+x^2} dx \\ &= \left[\frac{x^2}{2} (\arctan x)^2 \right]_0^1 - \int_0^1 \frac{x^2 \arctan x}{1+x^2} dx \\ &= \frac{1}{2} \left(\frac{\pi}{4} \right)^2 - \int_0^1 \frac{x^2 \arctan x}{1+x^2} dx \\ &= \frac{\pi^2}{32} - \int_0^1 \frac{x^2 \arctan x}{1+x^2} dx \end{aligned}$$

Now

$$\frac{x^2}{1+x^2} = 1 - \frac{1}{1+x^2}$$

so

$$I = \frac{\pi^2}{32} - \int_0^1 \arctan x dx + \int_0^1 \frac{\arctan x}{1+x^2} dx$$

We now evaluate these two integrals separately.

First, let

$$J = \int_0^1 \arctan x dx$$

Using integration by parts with

$$u = \arctan x, \quad \frac{dv}{dx} = 1$$

gives

$$\frac{du}{dx} = \frac{1}{1+x^2}, \quad v = x$$

So

$$\begin{aligned} J &= [x \arctan x]_0^1 - \int_0^1 \frac{x}{1+x^2} dx \\ &= \frac{\pi}{4} - \int_0^1 \frac{x}{1+x^2} dx \end{aligned}$$

Now use the substitution $t = 1 + x^2$, so $dt = 2x dx$. Then

$$\begin{aligned} \int_0^1 \frac{x}{1+x^2} dx &= \frac{1}{2} \int_1^2 \frac{1}{t} dt \\ &= \frac{1}{2} [\ln t]_1^2 \\ &= \frac{1}{2} \ln 2 \end{aligned}$$

Therefore

$$J = \frac{\pi}{4} - \frac{1}{2} \ln 2$$

Next, let

$$K = \int_0^1 \frac{\arctan x}{1+x^2} dx$$

Use the substitution

$$t = \arctan x$$

Then

$$x = \tan t, \quad dx = \sec^2 t \, dt, \quad 1 + x^2 = 1 + \tan^2 t = \sec^2 t$$

When $x = 0$, $t = 0$, and when $x = 1$, $t = \frac{\pi}{4}$. Hence

$$\begin{aligned} K &= \int_0^{\pi/4} \frac{t}{\sec^2 t} \sec^2 t \, dt \\ &= \int_0^{\pi/4} t \, dt \\ &= \left[\frac{t^2}{2} \right]_0^{\pi/4} \\ &= \frac{\pi^2}{32} \end{aligned}$$

Substitute J and K back into the expression for I :

$$\begin{aligned} I &= \frac{\pi^2}{32} - \left(\frac{\pi}{4} - \frac{1}{2} \ln 2 \right) + \frac{\pi^2}{32} \\ &= \frac{\pi^2}{16} - \frac{\pi}{4} + \frac{1}{2} \ln 2 \end{aligned}$$

Therefore

$$\int_0^1 x(\arctan x)^2 \, dx = \frac{\pi^2}{16} - \frac{\pi}{4} + \frac{1}{2} \ln 2$$

7. (a) Find

$$\int x^3 \arctan x \, dx \quad [5]$$

(b) Hence find the exact value of

$$\int_0^{\sqrt{3}} x^3 \arctan x \, dx \quad [2]$$

Solution

(a) Use integration by parts with

$$u = \arctan x \quad \frac{dv}{dx} = x^3$$

so that

$$\frac{du}{dx} = \frac{1}{1+x^2}, \quad v = \frac{x^4}{4}$$

Then

$$\begin{aligned} \int x^3 \arctan x \, dx &= \int u \frac{dv}{dx} \, dx \\ &= uv - \int v \frac{du}{dx} \, dx \\ &= \frac{x^4}{4} \arctan x - \int \frac{x^4}{4(1+x^2)} \, dx \\ &= \frac{x^4}{4} \arctan x - \frac{1}{4} \int \frac{x^4}{1+x^2} \, dx \end{aligned}$$

Now simplify the fraction by division:

$$\frac{x^4}{1+x^2} = x^2 - 1 + \frac{1}{1+x^2}$$

since

$$(1+x^2)(x^2-1) + 1 = x^4 - 1 + 1 = x^4$$

So

$$\begin{aligned} \int x^3 \arctan x \, dx &= \frac{x^4}{4} \arctan x - \frac{1}{4} \int \left(x^2 - 1 + \frac{1}{1+x^2} \right) \, dx \\ &= \frac{x^4}{4} \arctan x - \frac{1}{4} \left(\int x^2 \, dx - \int 1 \, dx + \int \frac{1}{1+x^2} \, dx \right) \\ &= \frac{x^4}{4} \arctan x - \frac{1}{4} \left(\frac{x^3}{3} - x + \arctan x \right) + C \\ &= \frac{x^4}{4} \arctan x - \frac{x^3}{12} + \frac{x}{4} - \frac{1}{4} \arctan x + C \\ &= \frac{x^4-1}{4} \arctan x - \frac{x^3}{12} + \frac{x}{4} + C \end{aligned}$$

Therefore,

$$\int x^3 \arctan x \, dx = \frac{x^4-1}{4} \arctan x - \frac{x^3}{12} + \frac{x}{4} + C$$

(b) Using the result from part (a), let

$$F(x) = \frac{x^4-1}{4} \arctan x - \frac{x^3}{12} + \frac{x}{4}$$

Then

$$\int_0^{\sqrt{3}} x^3 \arctan x \, dx = F(\sqrt{3}) - F(0)$$

First, at $x = \sqrt{3}$,

$$\arctan(\sqrt{3}) = \frac{\pi}{3}, \quad (\sqrt{3})^4 = 9, \quad (\sqrt{3})^3 = 3\sqrt{3}$$

so

$$\begin{aligned} F(\sqrt{3}) &= \frac{9-1}{4} \cdot \frac{\pi}{3} - \frac{3\sqrt{3}}{12} + \frac{\sqrt{3}}{4} \\ &= \frac{8}{4} \cdot \frac{\pi}{3} - \frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} \\ &= \frac{2\pi}{3} \end{aligned}$$

Also,

$$F(0) = \frac{0-1}{4} \arctan 0 - 0 + 0 = 0$$

Hence

$$\int_0^{\sqrt{3}} x^3 \arctan x \, dx = \frac{2\pi}{3}$$

So the exact value is $\frac{2\pi}{3}$.

8. (a) Using the logarithmic form

$$\operatorname{artanh} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right), \quad |x| < 1$$

prove that the derivative of $\operatorname{artanh} x$ is

$$\frac{1}{1-x^2}$$

[5]

(b) Hence find

$$\int_0^{1/3} \operatorname{artanh}(2x) \, dx$$

giving your answer in exact logarithmic form.

[5]

(c) Noah tries to evaluate

$$\int_0^{1/2} \operatorname{artanh}(2x) \, dx$$

using his calculator, and gets an error. Explain why.

[1]

Solution

(a) Using the given logarithmic form,

$$\operatorname{artanh} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right) = \frac{1}{2} [\ln(1+x) - \ln(1-x)]$$

for $|x| < 1$.

Differentiate term by term:

$$\begin{aligned} \frac{d}{dx}(\operatorname{artanh} x) &= \frac{1}{2} \left(\frac{d}{dx} \ln(1+x) - \frac{d}{dx} \ln(1-x) \right) \\ &= \frac{1}{2} \left(\frac{1}{1+x} - \frac{-1}{1-x} \right) \\ &= \frac{1}{2} \left(\frac{1}{1+x} + \frac{1}{1-x} \right) \\ &= \frac{1}{2} \left(\frac{(1-x) + (1+x)}{(1+x)(1-x)} \right) \\ &= \frac{1}{2} \left(\frac{2}{1-x^2} \right) \\ &= \frac{1}{1-x^2} \end{aligned}$$

Hence

$$\frac{d}{dx}(\operatorname{artanh} x) = \frac{1}{1-x^2}$$

(b) On $0 \leq x \leq \frac{1}{3}$, we have $0 \leq 2x \leq \frac{2}{3} < 1$, so $\operatorname{artanh}(2x)$ is defined.

Let

$$I = \int_0^{1/3} \operatorname{artanh}(2x) \, dx$$

Use integration by parts with

$$u = \operatorname{artanh}(2x), \quad \frac{dv}{dx} = 1$$

Then

$$v = x$$

and from part (a), using the chain rule,

$$\frac{du}{dx} = \frac{2}{1 - (2x)^2} = \frac{2}{1 - 4x^2}$$

So

$$\begin{aligned} I &= [x \operatorname{artanh}(2x)]_0^{1/3} - \int_0^{1/3} x \cdot \frac{2}{1 - 4x^2} dx \\ &= [x \operatorname{artanh}(2x)]_0^{1/3} - \int_0^{1/3} \frac{2x}{1 - 4x^2} dx \end{aligned}$$

For the remaining integral, let

$$t = 1 - 4x^2$$

Then

$$dt = -8x dx$$

so

$$2x dx = -\frac{1}{4} dt$$

When $x = 0$, $t = 1$, and when $x = \frac{1}{3}$, $t = \frac{5}{9}$. Therefore

$$\begin{aligned} \int_0^{1/3} \frac{2x}{1 - 4x^2} dx &= -\frac{1}{4} \int_1^{5/9} \frac{1}{t} dt \\ &= -\frac{1}{4} [\ln t]_1^{5/9} \\ &= -\frac{1}{4} \ln \left(\frac{5}{9} \right) \end{aligned}$$

Hence

$$\begin{aligned} I &= [x \operatorname{artanh}(2x)]_0^{1/3} + \frac{1}{4} \ln \left(\frac{5}{9} \right) \\ &= \frac{1}{3} \operatorname{artanh} \left(\frac{2}{3} \right) + \frac{1}{4} \ln \left(\frac{5}{9} \right) \end{aligned}$$

Now use the logarithmic form again:

$$\begin{aligned} \operatorname{artanh} \left(\frac{2}{3} \right) &= \frac{1}{2} \ln \left(\frac{1 + \frac{2}{3}}{1 - \frac{2}{3}} \right) \\ &= \frac{1}{2} \ln \left(\frac{5/3}{1/3} \right) \\ &= \frac{1}{2} \ln 5 \end{aligned}$$

So

$$\begin{aligned} I &= \frac{1}{3} \cdot \frac{1}{2} \ln 5 + \frac{1}{4} \ln \left(\frac{5}{9} \right) \\ &= \frac{1}{6} \ln 5 + \frac{1}{4} (\ln 5 - \ln 9) \\ &= \frac{1}{6} \ln 5 + \frac{1}{4} \ln 5 - \frac{1}{2} \ln 3 \\ &= \frac{5}{12} \ln 5 - \frac{1}{2} \ln 3 \end{aligned}$$

This can also be written as

$$\begin{aligned} I &= \frac{1}{12} (5 \ln 5 - 6 \ln 3) \\ &= \frac{1}{12} \ln \left(\frac{5^5}{3^6} \right) \\ &= \frac{1}{12} \ln \left(\frac{3125}{729} \right) \end{aligned}$$

Therefore

$$\int_0^{1/3} \operatorname{artanh}(2x) \, dx = \frac{5}{12} \ln 5 - \frac{1}{2} \ln 3 = \frac{1}{12} \ln \left(\frac{3125}{729} \right)$$

- (c) At the upper limit $x = \frac{1}{2}$, we have $2x = 1$, and $\operatorname{artanh}(1)$ is undefined because $\operatorname{artanh} t$ is only defined for $|t| < 1$.

So the integrand is not defined at the endpoint, and a calculator treating this as an ordinary definite integral gives an error.

9. For $0 < x < 1$, use integration by parts to find

$$\int x \operatorname{artanh}(2x - 1) \, dx$$

giving your final answer in terms of logarithms.

[6]

Solution

Let

$$I = \int x \operatorname{artanh}(2x - 1) \, dx$$

Use integration by parts with

$$u = \operatorname{artanh}(2x - 1), \quad \frac{dv}{dx} = x$$

Then

$$v = \frac{x^2}{2}$$

and

$$\begin{aligned} \frac{du}{dx} &= \frac{2}{1 - (2x - 1)^2} \\ &= \frac{2}{1 - (4x^2 - 4x + 1)} \\ &= \frac{2}{4x - 4x^2} \\ &= \frac{1}{2x(1 - x)} \end{aligned}$$

So, by integration by parts,

$$\begin{aligned} I &= uv - \int v \frac{du}{dx} \, dx \\ &= \frac{x^2}{2} \operatorname{artanh}(2x - 1) - \int \frac{x^2}{2} \cdot \frac{1}{2x(1 - x)} \, dx \\ &= \frac{x^2}{2} \operatorname{artanh}(2x - 1) - \frac{1}{4} \int \frac{x}{1 - x} \, dx \end{aligned}$$

Now simplify the fraction:

$$\frac{x}{1 - x} = -1 + \frac{1}{1 - x}$$

since

$$-1 + \frac{1}{1 - x} = \frac{-(1 - x) + 1}{1 - x} = \frac{x}{1 - x}$$

Hence

$$\begin{aligned} \int \frac{x}{1 - x} \, dx &= \int \left(-1 + \frac{1}{1 - x} \right) \, dx \\ &= -x - \ln(1 - x) \end{aligned}$$

because

$$\int \frac{1}{1 - x} \, dx = -\ln(1 - x)$$

Therefore

$$\begin{aligned} I &= \frac{x^2}{2} \operatorname{artanh}(2x - 1) - \frac{1}{4}(-x - \ln(1 - x)) \\ &= \frac{x^2}{2} \operatorname{artanh}(2x - 1) + \frac{x}{4} + \frac{1}{4} \ln(1 - x) + C \end{aligned}$$

Now write artanh in logarithmic form:

$$\operatorname{artanh} t = \frac{1}{2} \ln \left(\frac{1 + t}{1 - t} \right)$$

With $t = 2x - 1$,

$$\operatorname{artanh}(2x - 1) = \frac{1}{2} \ln \left(\frac{1 + (2x - 1)}{1 - (2x - 1)} \right) = \frac{1}{2} \ln \left(\frac{2x}{2 - 2x} \right) = \frac{1}{2} \ln \left(\frac{x}{1 - x} \right)$$

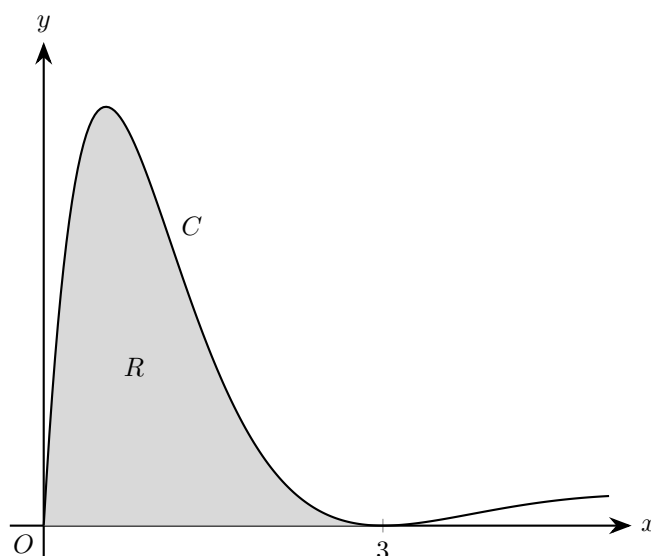
Since $0 < x < 1$, both x and $1 - x$ are positive, so these logarithms are valid.

Substitute this into the expression for I :

$$\begin{aligned} I &= \frac{x^2}{2} \cdot \frac{1}{2} \ln \left(\frac{x}{1 - x} \right) + \frac{x}{4} + \frac{1}{4} \ln(1 - x) + C \\ &= \frac{x^2}{4} (\ln x - \ln(1 - x)) + \frac{x}{4} + \frac{1}{4} \ln(1 - x) + C \\ &= \frac{x^2}{4} \ln x + \left(-\frac{x^2}{4} + \frac{1}{4} \right) \ln(1 - x) + \frac{x}{4} + C \\ &= \frac{x^2}{4} \ln x + \frac{1 - x^2}{4} \ln(1 - x) + \frac{x}{4} + C \end{aligned}$$

So,

$$\int x \operatorname{artanh}(2x - 1) \, dx = \frac{x^2}{4} \ln x + \frac{1 - x^2}{4} \ln(1 - x) + \frac{x}{4} + C$$



10. The sketch shows part of the curve C given by

$$y = x(3-x)^2 e^{-x} \quad x \geq 0$$

The shaded region R is bounded by C and the x -axis for $0 \leq x \leq 3$.

Find the exact area of R , giving your answer in the form

$$A + Be^{-3}$$

where A and B are rational numbers.

[5]

Solution

For $0 \leq x \leq 3$,

$$x \geq 0, \quad (3-x)^2 \geq 0, \quad e^{-x} > 0$$

so the curve lies above the x -axis, and the required area is

$$\int_0^3 x(3-x)^2 e^{-x} dx$$

First expand the bracket:

$$\begin{aligned} x(3-x)^2 &= x(9-6x+x^2) \\ &= 9x-6x^2+x^3 \end{aligned}$$

so

$$\text{Area} = \int_0^3 (x^3 - 6x^2 + 9x)e^{-x} dx$$

Now integrate by parts repeatedly, using the tabular method:

Sign	D	I
+	$x^3 - 6x^2 + 9x$	e^{-x}
-	$3x^2 - 12x + 9$	$-e^{-x}$
+	$6x - 12$	e^{-x}
-	6	$-e^{-x}$
+	0	e^{-x}

Reading the table diagonally gives

$$\begin{aligned}\int (x^3 - 6x^2 + 9x)e^{-x} dx &= -(x^3 - 6x^2 + 9x)e^{-x} - (3x^2 - 12x + 9)e^{-x} \\ &\quad - (6x - 12)e^{-x} - 6e^{-x} + C \\ &= (-x^3 + 3x^2 - 3x - 3)e^{-x} + C\end{aligned}$$

Therefore

$$\begin{aligned}\text{Area} &= [(-x^3 + 3x^2 - 3x - 3)e^{-x}]_0^3 \\ &= (-27 + 27 - 9 - 3)e^{-3} - ((-3)e^0) \\ &= -12e^{-3} + 3 \\ &= 3 - 12e^{-3}\end{aligned}$$

So the exact area is

$$3 - 12e^{-3}$$

which is of the form $A + Be^{-3}$ with

$$A = 3, \quad B = -12$$

11. Show that

$$\int_{e^{-1/2}}^1 x(\ln x)^2 dx = a + \frac{b}{e}$$

where a and b are rational constants to be found.

[6]

Solution

Let

$$I = \int_{e^{-1/2}}^1 x(\ln x)^2 dx$$

Use the substitution

$$u = \ln x$$

Then

$$x = e^u \quad \text{and} \quad dx = e^u du$$

So the integrand becomes

$$x(\ln x)^2 dx = e^u \cdot u^2 \cdot e^u du = u^2 e^{2u} du$$

Now change the limits.

When $x = e^{-1/2}$,

$$u = \ln(e^{-1/2}) = -\frac{1}{2}$$

When $x = 1$,

$$u = \ln 1 = 0$$

Hence

$$I = \int_{-1/2}^0 u^2 e^{2u} du$$

To integrate this, use integration by parts twice.

First,

$$\int u^2 e^{2u} du$$

Take

$$p = u^2, \quad \frac{dq}{du} = e^{2u}$$

Then

$$\frac{dp}{du} = 2u, \quad q = \frac{1}{2}e^{2u}$$

So

$$\int u^2 e^{2u} du = \frac{1}{2}u^2 e^{2u} - \int u e^{2u} du$$

Now integrate $\int u e^{2u} du$ by parts again.

Take

$$p = u, \quad \frac{dq}{du} = e^{2u}$$

Then

$$\frac{dp}{du} = 1, \quad q = \frac{1}{2}e^{2u}$$

Hence

$$\int u e^{2u} du = \frac{1}{2}u e^{2u} - \frac{1}{2} \int e^{2u} du = \frac{1}{2}u e^{2u} - \frac{1}{4}e^{2u}$$

Substituting back,

$$\begin{aligned} \int u^2 e^{2u} du &= \frac{1}{2}u^2 e^{2u} - \left(\frac{1}{2}u e^{2u} - \frac{1}{4}e^{2u} \right) \\ &= e^{2u} \left(\frac{u^2}{2} - \frac{u}{2} + \frac{1}{4} \right) \end{aligned}$$

Therefore

$$I = \left[e^{2u} \left(\frac{u^2}{2} - \frac{u}{2} + \frac{1}{4} \right) \right]_{-1/2}^0$$

Now substitute the limits:

$$\begin{aligned} I &= e^0 \left(\frac{0^2}{2} - \frac{0}{2} + \frac{1}{4} \right) - e^{-1} \left(\frac{(-1/2)^2}{2} - \frac{-1/2}{2} + \frac{1}{4} \right) \\ &= \frac{1}{4} - \frac{1}{e} \left(\frac{1}{8} + \frac{1}{4} + \frac{1}{4} \right) \\ &= \frac{1}{4} - \frac{1}{e} \left(\frac{5}{8} \right) \\ &= \frac{1}{4} - \frac{5}{8e} \end{aligned}$$

So

$$\int_{1/\sqrt{e}}^1 x(\ln x)^2 dx = \frac{1}{4} - \frac{5}{8e}$$

Hence it is of the form $a + \frac{b}{e}$ with

$$a = \frac{1}{4}, \quad b = -\frac{5}{8}$$

12. (a) Using the logarithmic form of $\operatorname{artanh} x$, prove that the derivative of $\operatorname{artanh} x$ is

$$\frac{1}{1-x^2}$$

for $|x| < 1$.

[5]

- (b) Hence find

$$\int_0^{1/2} \operatorname{artanh} x \, dx$$

giving your answer in exact logarithmic form.

[5]

- (c) Priya tries to evaluate

$$\int_0^{3/2} \operatorname{artanh} x \, dx$$

using her calculator, and gets an 'error'. Explain why.

[1]

Solution

- (a) For $|x| < 1$,

$$\operatorname{artanh} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right) = \frac{1}{2} [\ln(1+x) - \ln(1-x)]$$

This is valid for $|x| < 1$ because we need $1+x > 0$ and $1-x > 0$.

Differentiate term by term:

$$\begin{aligned} \frac{d}{dx}(\operatorname{artanh} x) &= \frac{1}{2} \left[\frac{d}{dx} \ln(1+x) - \frac{d}{dx} \ln(1-x) \right] \\ &= \frac{1}{2} \left[\frac{1}{1+x} - \frac{1}{1-x}(-1) \right] \\ &= \frac{1}{2} \left[\frac{1}{1+x} + \frac{1}{1-x} \right] \\ &= \frac{1}{2} \left[\frac{(1-x) + (1+x)}{(1+x)(1-x)} \right] \\ &= \frac{1}{2} \left[\frac{2}{1-x^2} \right] \\ &= \frac{1}{1-x^2} \end{aligned}$$

Hence

$$\frac{d}{dx}(\operatorname{artanh} x) = \frac{1}{1-x^2} \quad (|x| < 1)$$

- (b) Using integration by parts with

$$u = \operatorname{artanh} x, \quad \frac{dv}{dx} = 1$$

we have

$$\frac{du}{dx} = \frac{1}{1-x^2}, \quad v = x$$

So

$$\begin{aligned} \int \operatorname{artanh} x \, dx &= x \operatorname{artanh} x - \int x \cdot \frac{1}{1-x^2} \, dx \\ &= x \operatorname{artanh} x - \int \frac{x}{1-x^2} \, dx \end{aligned}$$

Now let $t = 1 - x^2$, so

$$\frac{dt}{dx} = -2x \quad \Rightarrow \quad dt = -2x \, dx$$

Then

$$\begin{aligned}\int \frac{x}{1-x^2} dx &= -\frac{1}{2} \int \frac{1}{t} dt \\ &= -\frac{1}{2} \ln t + C \\ &= -\frac{1}{2} \ln(1-x^2) + C\end{aligned}$$

Therefore

$$\int \operatorname{artanh} x \, dx = x \operatorname{artanh} x + \frac{1}{2} \ln(1-x^2) + C$$

Hence

$$\begin{aligned}\int_0^{1/2} \operatorname{artanh} x \, dx &= \left[x \operatorname{artanh} x + \frac{1}{2} \ln(1-x^2) \right]_0^{1/2} \\ &= \frac{1}{2} \operatorname{artanh} \left(\frac{1}{2} \right) + \frac{1}{2} \ln \left(\frac{3}{4} \right) - \left(0 + \frac{1}{2} \ln 1 \right)\end{aligned}$$

Now

$$\operatorname{artanh} \left(\frac{1}{2} \right) = \frac{1}{2} \ln \left(\frac{1+\frac{1}{2}}{1-\frac{1}{2}} \right) = \frac{1}{2} \ln 3$$

So

$$\begin{aligned}\int_0^{1/2} \operatorname{artanh} x \, dx &= \frac{1}{2} \cdot \frac{1}{2} \ln 3 + \frac{1}{2} \ln \left(\frac{3}{4} \right) \\ &= \frac{1}{4} \ln 3 + \frac{1}{2} (\ln 3 - \ln 4) \\ &= \frac{1}{4} \ln 3 + \frac{1}{2} \ln 3 - \ln 2 \\ &= \frac{3}{4} \ln 3 - \ln 2\end{aligned}$$

Also,

$$\frac{3}{4} \ln 3 - \ln 2 = \frac{1}{4} \ln \left(\frac{27}{16} \right)$$

Therefore

$$\int_0^{1/2} \operatorname{artanh} x \, dx = \frac{3}{4} \ln 3 - \ln 2 = \frac{1}{4} \ln \left(\frac{27}{16} \right)$$

- (c) In the real number system, $\operatorname{artanh} x$ is defined only for $|x| < 1$. The interval $[0, \frac{3}{2}]$ includes $x = 1$, where $\operatorname{artanh} x$ is undefined, and values $x > 1$, where it is not real. So the calculator gives a domain error.

13. Given that

$$y = \arcsin\left(\frac{x-1}{3}\right)$$

(a) find $\frac{dy}{dx}$, giving your answer in its simplest form. [2]

(b) Use integration by parts to find

$$\int \arcsin\left(\frac{x-1}{3}\right) dx \quad [4]$$

Solution

(a) Let

$$u = \frac{x-1}{3}$$

Then

$$y = \arcsin u$$

so by the chain rule,

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{\sqrt{1-u^2}} \cdot \frac{du}{dx} \\ &= \frac{1}{\sqrt{1-\left(\frac{x-1}{3}\right)^2}} \cdot \frac{1}{3} \\ &= \frac{1}{3\sqrt{1-\frac{(x-1)^2}{9}}} \\ &= \frac{1}{\sqrt{9-(x-1)^2}} \end{aligned}$$

Hence

$$\frac{dy}{dx} = \frac{1}{\sqrt{9-(x-1)^2}}$$

(b) Let

$$I = \int \arcsin\left(\frac{x-1}{3}\right) dx$$

Use integration by parts with

$$u = \arcsin\left(\frac{x-1}{3}\right), \quad \frac{dv}{dx} = 1$$

Then

$$v = x$$

and from part (a),

$$\frac{du}{dx} = \frac{1}{\sqrt{9-(x-1)^2}}$$

So

$$\begin{aligned} I &= uv - \int v \frac{du}{dx} dx \\ &= x \arcsin\left(\frac{x-1}{3}\right) - \int \frac{x}{\sqrt{9-(x-1)^2}} dx \end{aligned}$$

Now write

$$x = (x-1) + 1$$

so

$$\int \frac{x}{\sqrt{9 - (x - 1)^2}} dx = \int \frac{x - 1}{\sqrt{9 - (x - 1)^2}} dx + \int \frac{1}{\sqrt{9 - (x - 1)^2}} dx$$

For the first integral, let

$$t = 9 - (x - 1)^2$$

Then

$$\frac{dt}{dx} = -2(x - 1) \quad \Rightarrow \quad (x - 1) dx = -\frac{1}{2} dt$$

Hence

$$\begin{aligned} \int \frac{x - 1}{\sqrt{9 - (x - 1)^2}} dx &= -\frac{1}{2} \int t^{-1/2} dt \\ &= -t^{1/2} \\ &= -\sqrt{9 - (x - 1)^2} \end{aligned}$$

For the second integral, using the result from part (a) in reverse,

$$\int \frac{1}{\sqrt{9 - (x - 1)^2}} dx = \arcsin\left(\frac{x - 1}{3}\right)$$

Therefore

$$\begin{aligned} I &= x \arcsin\left(\frac{x - 1}{3}\right) - \left(-\sqrt{9 - (x - 1)^2} + \arcsin\left(\frac{x - 1}{3}\right)\right) + C \\ &= x \arcsin\left(\frac{x - 1}{3}\right) + \sqrt{9 - (x - 1)^2} - \arcsin\left(\frac{x - 1}{3}\right) + C \\ &= (x - 1) \arcsin\left(\frac{x - 1}{3}\right) + \sqrt{9 - (x - 1)^2} + C \end{aligned}$$

Hence

$$\int \arcsin\left(\frac{x - 1}{3}\right) dx = (x - 1) \arcsin\left(\frac{x - 1}{3}\right) + \sqrt{9 - (x - 1)^2} + C$$

14. Use integration by parts to find

$$\int_1^9 \sqrt{x} \ln\left(\frac{9}{x}\right) dx$$

giving your answer in the form $a + b \ln 3$, where a and b are constants to be found.

[5]

Solution

Let

$$u = \ln\left(\frac{9}{x}\right), \quad \frac{dv}{dx} = \sqrt{x}$$

Then

$$\frac{du}{dx} = -\frac{1}{x}, \quad v = \int x^{1/2} dx = \frac{2}{3}x^{3/2}$$

Using integration by parts,

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

So if

$$I = \int_1^9 \sqrt{x} \ln\left(\frac{9}{x}\right) dx$$

then

$$\begin{aligned} I &= \left[\frac{2}{3}x^{3/2} \ln\left(\frac{9}{x}\right) \right]_1^9 - \int_1^9 \frac{2}{3}x^{3/2} \left(-\frac{1}{x}\right) dx \\ &= \left[\frac{2}{3}x^{3/2} \ln\left(\frac{9}{x}\right) \right]_1^9 + \frac{2}{3} \int_1^9 x^{1/2} dx \end{aligned}$$

Now evaluate the first term:

$$\begin{aligned} \left[\frac{2}{3}x^{3/2} \ln\left(\frac{9}{x}\right) \right]_1^9 &= \frac{2}{3}(9)^{3/2} \ln\left(\frac{9}{9}\right) - \frac{2}{3}(1)^{3/2} \ln\left(\frac{9}{1}\right) \\ &= \frac{2}{3}(27) \ln 1 - \frac{2}{3} \ln 9 \\ &= 0 - \frac{2}{3}(2 \ln 3) \\ &= -\frac{4}{3} \ln 3 \end{aligned}$$

Now evaluate the integral:

$$\begin{aligned} \frac{2}{3} \int_1^9 x^{1/2} dx &= \frac{2}{3} \left[\frac{2}{3}x^{3/2} \right]_1^9 \\ &= \frac{4}{9} \left((9)^{3/2} - 1^{3/2} \right) \\ &= \frac{4}{9}(27 - 1) \\ &= \frac{104}{9} \end{aligned}$$

Hence

$$I = \frac{104}{9} - \frac{4}{3} \ln 3$$

So in the form $a + b \ln 3$,

$$a = \frac{104}{9}, \quad b = -\frac{4}{3}$$

15.

$$g(\theta) = (1 + \sin \theta - \cos \theta)^2$$

(a) Show that $g(\theta) = 2 + 2 \sin \theta - 2 \cos \theta - \sin 2\theta$. [2]

(b) Hence, using calculus, find the exact value of

$$\int_0^{\pi/2} \theta g(\theta) \, d\theta \quad [5]$$

Solution

(a) Expanding the square gives

$$\begin{aligned} g(\theta) &= (1 + \sin \theta - \cos \theta)^2 \\ &= 1 + \sin^2 \theta + \cos^2 \theta + 2 \sin \theta - 2 \cos \theta - 2 \sin \theta \cos \theta \end{aligned}$$

Using $\sin^2 \theta + \cos^2 \theta = 1$ and $\sin 2\theta = 2 \sin \theta \cos \theta$,

$$\begin{aligned} g(\theta) &= 1 + 1 + 2 \sin \theta - 2 \cos \theta - 2 \sin \theta \cos \theta \\ &= 2 + 2 \sin \theta - 2 \cos \theta - \sin 2\theta \end{aligned}$$

Hence

$$g(\theta) = 2 + 2 \sin \theta - 2 \cos \theta - \sin 2\theta$$

(b) Let

$$I = \int_0^{\pi/2} \theta g(\theta) \, d\theta$$

Using part (a),

$$\begin{aligned} I &= \int_0^{\pi/2} \theta (2 + 2 \sin \theta - 2 \cos \theta - \sin 2\theta) \, d\theta \\ &= 2 \int_0^{\pi/2} \theta \, d\theta + 2 \int_0^{\pi/2} \theta \sin \theta \, d\theta - 2 \int_0^{\pi/2} \theta \cos \theta \, d\theta - \int_0^{\pi/2} \theta \sin 2\theta \, d\theta \end{aligned}$$

Now evaluate each integral.

First,

$$\begin{aligned} 2 \int_0^{\pi/2} \theta \, d\theta &= 2 \left[\frac{\theta^2}{2} \right]_0^{\pi/2} \\ &= [\theta^2]_0^{\pi/2} \\ &= \frac{\pi^2}{4} \end{aligned}$$

Next, for $\int_0^{\pi/2} \theta \sin \theta \, d\theta$, use integration by parts with $u = \theta$, $\frac{dv}{d\theta} = \sin \theta$

Then $\frac{du}{d\theta} = 1$ and $v = -\cos \theta$, so

$$\begin{aligned} \int_0^{\pi/2} \theta \sin \theta \, d\theta &= [-\theta \cos \theta]_0^{\pi/2} + \int_0^{\pi/2} \cos \theta \, d\theta \\ &= [-\theta \cos \theta + \sin \theta]_0^{\pi/2} \\ &= 1 \end{aligned}$$

For $\int_0^{\pi/2} \theta \cos \theta \, d\theta$, again use integration by parts with $u = \theta$, $\frac{dv}{d\theta} = \cos \theta$

Then $\frac{du}{d\theta} = 1$ and $v = \sin \theta$, so

$$\begin{aligned}\int_0^{\pi/2} \theta \cos \theta \, d\theta &= [\theta \sin \theta]_0^{\pi/2} - \int_0^{\pi/2} \sin \theta \, d\theta \\ &= [\theta \sin \theta + \cos \theta]_0^{\pi/2} \\ &= \frac{\pi}{2} - 1\end{aligned}$$

For $\int_0^{\pi/2} \theta \sin 2\theta \, d\theta$, use integration by parts with $u = \theta$, $\frac{dv}{d\theta} = \sin 2\theta$

Then $\frac{du}{d\theta} = 1$ and $v = -\frac{1}{2} \cos 2\theta$, so

$$\begin{aligned}\int_0^{\pi/2} \theta \sin 2\theta \, d\theta &= \left[-\frac{1}{2}\theta \cos 2\theta\right]_0^{\pi/2} + \frac{1}{2} \int_0^{\pi/2} \cos 2\theta \, d\theta \\ &= \left[-\frac{1}{2}\theta \cos 2\theta + \frac{1}{4} \sin 2\theta\right]_0^{\pi/2} \\ &= \frac{\pi}{4}\end{aligned}$$

Substituting these into I ,

$$\begin{aligned}I &= \frac{\pi^2}{4} + 2(1) - 2\left(\frac{\pi}{2} - 1\right) - \frac{\pi}{4} \\ &= \frac{\pi^2}{4} + 2 - \pi + 2 - \frac{\pi}{4} \\ &= \frac{\pi^2}{4} - \frac{5\pi}{4} + 4 \\ &= \frac{\pi^2 - 5\pi + 16}{4}\end{aligned}$$

Therefore,

$$\int_0^{\pi/2} \theta g(\theta) \, d\theta = \frac{\pi^2 - 5\pi + 16}{4}$$